

AUTOMATION TESTING WITH SELENIUM - COURSE

MODULE

❖ Module 1: Introduction to Automation Testing

- What is Automation Testing?
 - Manual vs Automation Testing
 - Benefits of Automation
 - When to Automate?
 - Overview of Popular Automation Tools
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❖ Module 2: Java Basics for Testers

- Installing JDK and IDE (Eclipse/IntelliJ)
 - Variables, Data Types, Operators
 - Control Statements (if, switch, loops)
 - Arrays and Strings
 - OOPs Concepts (Class, Object, Inheritance, Polymorphism, etc.)
 - Exception Handling
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❖ Module 3: Selenium WebDriver Introduction

- What is Selenium?
 - Selenium Components: WebDriver, IDE, Grid
 - Browser Architecture and Driver Setup
 - Launching Browsers using WebDriver
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❖ Module 4: Locators and WebElements

- Types of Locators (ID, Name, ClassName, XPath, CSS Selector, etc.)
 - Finding Elements
 - WebElement Methods (click, sendKeys, getText, isDisplayed, etc.)
 - XPath and CSS Selector Deep Dive
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❖ **Module 5: Handling Web Controls**

- Dropdowns (Select class)
 - Checkboxes and Radio Buttons
 - Alerts and Popups (Alert interface)
 - Frames and iFrames
 - Windows Handling (Switching between tabs/windows)
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❖ **Module 6: Waits and Synchronization**

- Implicit Wait
 - Explicit Wait
 - Fluent Wait
 - Thread.sleep vs WebDriverWait
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❖ **Module 7: TestNG Framework**

- What is TestNG and Why Use It?
 - TestNG Annotations
 - TestNG XML Configuration
 - Grouping, Prioritization, and Parallel Execution
 - Assertions and Reports
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❖ **Module 8: Automation Framework Design**

- What is a Framework?
 - Types: Modular, Data Driven, Keyword Driven, Hybrid
 - Folder Structure
 - Page Object Model (POM)
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❖ **Module 09: API Testing**

- What is API Testing?
 - How to test API
 - Postman Usage
 - API HTTP codes
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❖ **Module 10: Selenium Grid (Optional)**

- What is Selenium Grid?
 - Local vs Remote Execution
 - Parallel Cross-Browser Testing
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❖ **Module 11: Git Integration**

- Git Basics: clone, commit, push, pull
 - Integrating Selenium with Git
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❖ **Module 12: CI/CD Introduction with Jenkins**

- Jenkins Overview
 - Creating and Running a Job
 - Integrating Git with Jenkins
 - Scheduling and Automating Test Runs
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❖ **Bonus: Real-Time Project + Interview Preparation**

- End-to-End Project: Build a mini automation suite
 - Resume Preparation for Automation Tester
 - Common Interview Questions
 - Mock Interview/Code Review
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❖ **Tools Covered:**

- Java ()
- Selenium WebDriver
- TestNG
- Git and Github
- Jenkins
- Postman
- Excel / Data Files
- Browser Drivers (ChromeDriver)